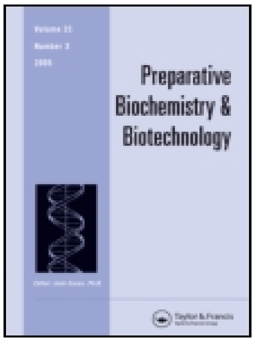

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Continuous Processing for Production of Biopharmaceuticals

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The merits of continuous processing over batch processing are well known in the manufacturing industry. Continuous operation results in shorter process times due to omission of hold steps, higher productivity due to reduced shutdown costs, and lowers labor requirement. Over the past decade, there has been an increasing interest in continuous processing within the bioprocessing community, specifically those involved in production of biotherapeutics. Continuous operations in upstream processing (perfusion) have been performed for decades. However, recent development of continuous downstream operations has led the industry to envisage an integrated bioprocessing platform for efficient production. The regulators, key players in the biotherapeutic industry, have also expressed their interest and willingness in this migration from the traditional batch processing. This paper aims to review major developments in continuous bioprocessing in the past decade. A discussion of pros and cons of the different proposed approaches has also been presented.

Keywords *bioprocessing, biotherapeutics, continuous chromatography, continuous processing, continuous refolding*

INTRODUCTION

Continuous bioprocessing utilizes a continuous flow of material through the various unit operations such that at steady state, product of consistent quality is being produced as long as the operation runs.[1] In batch manufacturing, material moves discretely through each unit operation and the product is produced in lots (or batches). Continuous manufacturing has long been adopted by many manufacturing sectors, such as steel casting, petrochemical, chemical, food, and pharmaceutical, but its implementation in biotechnology manufacturing (particularly of biotherapeutics) is still awaited. Exceptions to this include production of bioethanol and microalgae. Reasons for the relatively slow adoption of continuous bioprocessing in manufacturing of biotherapeutics include the sector's low tolerance to risk, concerns about implementing new technologies, financial investment required for successful implementation, and last but not the least regulatory implications of the increased complexity.[2,3]

More recently, the biotech industry has shown an increasing interest in continuous processing. Major regulators such as the U.S. Food and Drug Administration (FDA) have also

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expressed their belief in continuous processing, as it better aligns with the desire to directly build in quality into process design.[4] Further, both the U.S. FDA and European Medicines Agency (EMA) have also defined a batch through quantity of material rather than the mode of manufacturing (Code of Federal Regulations: 21 CFR 210.3). This reflects the regulatory acknowledgement of the concepts behind continuous manufacturing.

The major benefits of making the switch from batch processing to continuous processing include reduced costs, increased productivity, improved quality, and increased manufacturing flexibility.[5] Continuous operations allow production at significantly small scales, thereby resulting in lower capital cost, smaller footprint, higher automation, and finally, lower labor cost. From a scientific perspective, continuous operation could also be more controllable and thus result in better consistency in process and product quality.[6]

The efforts on implementation of continuous processing in the biotech industry have so far primarily focused on upstream processing. Specifically, use of a continuous (perfusion) reactor for high-cell-density cultures has been demonstrated extensively.[2,7] Researchers have achieved significant improvements in cell banking, inoculum expansion, and protein production to make this a feasible alternative. Chromatography, a major contributor to downstream processing, has also been targeted with limited success via annular chromatography[8] and simulated moving bed chromatography.[9] However, the true potential of continuous production can only be realized after successful integration of all unit operations of the process.

In this article, we review the work that has been done on this topic thus far and discuss the challenges. We will also compare the different continuous technologies that have been proposed thus far.

UPSTREAM PROCESSING

Continuous upstream processing, commonly called perfusion, has been implemented and explored in the biotech industry for decades. Sterile medium feed is fed into the continuous stirred tank bioreactor, from which an outlet stream is withdrawn continuously. In cases where culture medium is perfused at a required dilution rate that is less than cell growth rate, the cell growth needs to be controlled (as in a Chemostat or Turbidostat). In the remaining cases, where dilution rate is higher than the cell growth rate, cells need to be retained in the bioreactor and this is often achieved via continuous filtration (Figure 1), such as alternating tangential flow filtration (ATF).[10,11]

Single-Use Bioreactors

Single-use technologies have been of great interest within the bioprocessing community, particularly due to their implication on continuous bioprocessing.[12] For perfusion cell culture, single-use and hybrid perfusion-capable reactors, probes, and fluidics components are commercially available. Prime benefits of single-use technologies are lower capital and utility costs and elimination of cleaning and sterilization steps.[12]

Continuous-Flow Micro-Bioreactors

In a recent development, microscale continuous bioreactors have been examined for production of biopharmaceuticals.[13] The study concluded that productivity of the process developed at

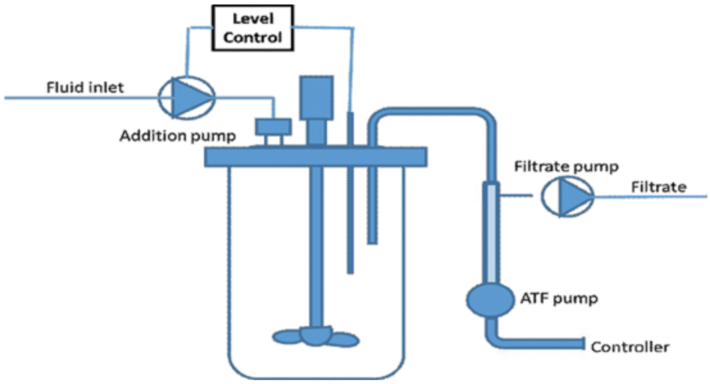

FIGURE 1 Continuous upstream processing, bioreactor coupled with ATF provides continuous removal of product from bioreactor with sterile medium feed continuously.[14]

microscale is three orders of magnitude higher than the traditional stirred tank fed batch bioreactors. The applicability of the technology, however, requires the cells to have strong binding to the device surface. This technology can be a significant enabler for creating a highly productive continuous bioprocessing train using a large number of such units operating in parallel. Other significant benefits that arise from operating at smaller scale include more effective process control due to homogeneous conditions and faster response.

DOWNSTREAM PROCESSING

The key objectives of downstream processing are clarification of the process stream, removal of process and host cell-related impurities, removal of product-related impurities, and finally, prevention of product degradation so as to achieve high recovery and satisfactory product quality profile. The primary unit operations involved include centrifugation, homogenization, filtration, refolding, chromatography, membrane separations, extraction, and precipitation. These unit operations are presently run in a batch fashion with the output of each step collected in a tank. In order to perform the entire process in the continuous mode, all the unit operations need to be integrated with the capability of recycling of streams and purging of impurities as required.

Continuous Cell Lysis

Cell lysis is required when protein expression is intracellular, as is the case with several microorganisms (such as *Escherichia coli*). Several methods are commonly used to physically lyse cells, including mechanical disruption, liquid homogenization, high-frequency sound waves, freeze/thaw cycles, manual grinding, and other enzymatic methods. Of these methods, high-pressure homogenization is not only the most popular but is also amenable for operation in a continuous fashion. In this unit operation, the cell suspension is forced at pressures of several hundred bars through an adjustable discharge valve with a restricted orifice. Cells are then disrupted by a combination of the high shear imposed, cavitation, the high-pressure drop, and the impact forces in the discharge valve (impingement). The homogenizer is operated either by using discrete multiple passes or continuously by use of a recycle loop to effect the multiple passes needed.[15–18] Zhu et al. have proposed a continuous thermal lysis protocol for large-scale preparation of plasmid DNA, where an optimized lysis buffer and a continuous thermal treatment with minimized number of centrifugation steps have been used to provide a scalable, efficient, and cost-effective plasmid preparation procedure for industrial production.[19]

It can be concluded that performing continuous cell lysis is quite feasible today.

Continuous Centrifugation

Centrifugation is a commonly used unit operation for solid–liquid separation. Of the two most commonly used designs, namely, tubular centrifuge (Figure 2a) and disk-stack centrifuge, the disk-stack centrifuge can be run in a pseudo-continuous manner with the suspensions fed and the clarified liquid removed in a continuous fashion.[5,7] The type of feed affects the choice of design with the split bowl used for high solid content feed such as yeast fermentation and disc nozzle for low solid content feeds.[5,20] Various other designs have also been proposed for continuous centrifugation, including hybrid centrifuge rotor,[21] continuous tubular bowl centrifuge,[22] and combined centrifugation and cell culture.[23]

Thus, it can be concluded that continuous centrifugation is also feasible and has been in practice over the past decades.

Continuous Precipitation

In cases of extracellular expression of recombinant proteins, precipitation offers an economical alternative for capturing the product from the multitude of components present in the feed stream. Typical procedure involves adding a precipitant to destabilize the solution resulting in formation of product-containing insoluble particles. The precipitant is usually specific to the product, with ionic precipitants used commonly. Further, mixing and shear characteristics of

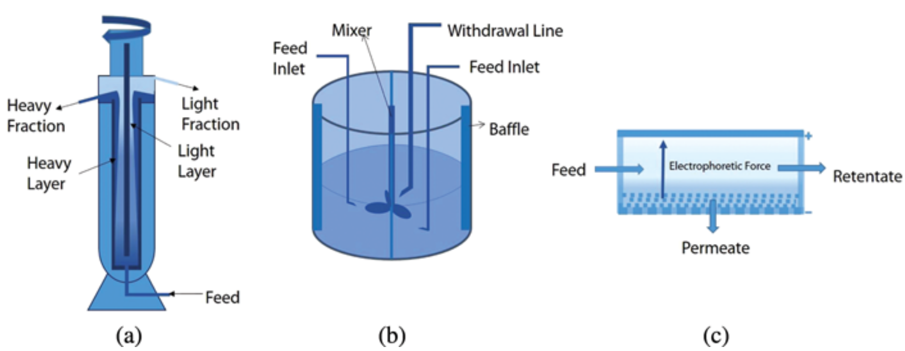

FIGURE 2 Schematic representations of various continuous downstream unit operations: (a) tubular continuous centrifuge, (b) mixed suspension and mixed product removal reactor (MSMPR) for continuous precipitation, and (c) continuous tangential flow electrofiltration.[32,33]

the unit are critical for its performance. Continuous precipitation has been performed in a variety of configurations. Mixed suspension and mixed product removal reactors (MSMPR) (Figure 2b) have been extensively investigated.[24,25] This reactor is a continuously stirred tank reactor (CSTR), which is continuously fed with protein solution and the precipitant while the processed stream is withdrawn. Precipitation has also been studied in a tubular reactor,[26,27] although the stability of flocks is a concern as compared to the stirred tank.[28] A more complex procedure, centrifugal precipitation chromatography, has also been used for continuous operation.[29]

Perhaps due to the complexity of the unit operation and its perceived deleterious impact on product stability, continuous precipitation has yet to be widely adopted in the bioprocessing industry.

Continuous Filtration

Filtration is arguably the simplest and thus the most commonly used unit operation for achieving solid–liquid separation. The application could involve harvest of cells, removal of by-products, concentration of a recombinant protein, or removal of viruses. Filtration can be classified depending upon the pore size of the filter used.[30] Microfiltration utilizes filters with pore sizes between 0.1 and 10 μm that allow water, salts, and macromolecules to flow through and retain suspended particles. Applications include cell harvesting and reagent preparation for analytical methods such as liquid chromatography. Ultrafiltration/diafiltration involves filters of pore sizes between 0.01 and 0.1 μm that allow permeation of water and salts but retain the macromolecules. These are often used for protein concentration and buffer exchange during processing. Finally, nanofiltration involves use of filters with pore size around 0.001 μm and is commonly used for virus removal. Filtration can also be complemented by electrophoretic techniques traditionally used for protein purification,[31] commonly called electrofiltration (Figure 2c).[32] The electric field can help in preventing or reducing the cake formation, thus making filtration more efficient.[33]

Continuous filtration has been attempted with both dead-end and tangential flow filtration formats. Single-pass dead-end filtration has been used for continuous filtration of solubilized inclusion bodies during refolding.[34] In this format gradual collection of particles occurs over time and a filter cake is formed.[35] This results in increase in the effective resistance of the filter, thus making filtration less efficient and requiring continuous removal of the cake (e.g., via reverse flushing). In this respect, tangential flow filtration is more amenable for continuous operation. Single-pass tangential flow filtration (SPTFF) is designed such that a single pass of solution is sufficient for achieving the desired filtration.[36] Another alternative to SPTFF involves recycle of the retentate and mixing it with fresh feed or use of several filtration units in a cascade configuration. For the case of diafiltration, cascade configuration has been attempted both in cocurrent and countercurrent configurations, with the latter found to be more efficient.[5,37] Alternating tangential flow filtration (ATF) has also been demonstrated as a successful process step for continuous capturing of the cells so as to enable perfusion cell culture operations.[11] ATF has been shown to offer near-linear scalability, low shear, and self-cleaning back flushing.

In conclusion, it can be said that enough alternatives exist to enable continuous filtration in bioprocessing.

Continuous Refolding

Recombinant proteins are often overexpressed in microbial systems such as *E. coli*, thereby resulting in formation of insoluble inclusion bodies.[38–40] This results in denaturation of the product in these inclusion bodies and requires refolding in order to attain the native structure. Typical refolding processes in industry are batch processes and involve dilution of solubilized inclusion bodies so as to simultaneously reduce the concentration of the denaturing agents and the protein triggering it to refold and avoid aggregation.[41] The degree of dilution in such methods is generally high, resulting in dilute protein concentrations ($<100\,\mu l/min$) and requirement of very large refold vessels.[5]

Mixing has been known to significantly impact efficiency of protein refolding, particularly during the addition of the denatured proteins to the refolding vessel at large scale.[42] Poor initial mixing can lead to higher aggregation due to high concentration of intermediates. Insufficient mixing can also lead in precipitation of proteins, which can be avoided by using dynamic inline mixing.[34]

A number of studies have targeted implementation of continuous refolding. Use of inline static mixing units for "fast mixing" and improving refolding performance has been reported.[43] A packed column plug flow reactor, supplemented by an initial mixing unit, has been shown to yield the efficiency of a batch reactor.[44] This configuration offers the additional flexibility of fine-tuning the refolding environment (temperature, reagent addition, etc.) to its optimal values at specific stages during refolding, thus resulting in a narrower distribution of the degree of refolding at the outlet. Fed batch addition of denatured protein has also been employed using a ceramic membrane tube to ensure gradual addition and mixing of the denatured protein into the flow reactor.[45] In this case, refolding efficiencies were found to be higher than the batch dilution method, as the fed batch addition significantly reduced the tendency of aggregate formation. In another configuration, a CSTR coupled to a diafiltration unit for removal of denaturants.[46] Recycling of unfolded protein has also been suggested. This configuration resulted in better cumulative yields than a batch reactor. In an improved version, the inclusion bodies could also be solubilized in a continuous manner before the protein solution is diluted with the refolding buffer.[43] Continuous refolding has also been performed in a tubular reactor with higher overall productivity than the batch reactor and with no requirement of emptying and refilling of reactor in continuous mode.[47] In an improved development, the tubular reactor was used for integrated continuous dissolution, refolding, and precipitation,[27] implying the flexibility of the configuration.

On-column refolding has also been attempted by many researchers in the past decade with the promise of simultaneous refolding and purification in a single step at high protein concentrations and yields.[48,49] Innovative designs of chromatographic equipment that allow for continuous refolding include preparative continuous annular chromatography (P-CAC) and simulated moving bed (SMB) chromatography. P-CAC involves continuous feeding of the denatured protein along with the refolding buffer onto a rotating annular column. The refolded protein, aggregates, and other impurities collect and elute at different angles at the bottom of the annular column. The elution profiles for a size-exclusion-based P-CAC have been shown to be similar to that from batch operation.[50] Since the aggregates are continuously separated from the product and are recycled back to the feed stream, the refolding yield is high.[51,52] A size-exclusion-based four-zone SMB chromatography has also been configured to enable on-column refolding of

FIGURE 3 Illustration of the different process schemes that enable continuous protein refolding.

the protein continuously with high efficiency and with low consumption of refolding buffer.[53] Continuous refolding has recently been performed in an assembly that uses a pipe reactor and an expanded bed adsorption column.[54] The denatured stream was diluted in a pipe reactor, after which the nascent refolded protein was sent onto the adsorption column, through which the concentrated and purified product was eluted continuously.

In summary, while on-column refolding needs more development before it can be commercialized, other configurations proposed in the preceding can be used for performing continuous refolding (Figure 3).

Continuous Chromatography

Chromatography plays a vital role in production of biotech therapeutics in that it forms the backbone of the purification process due to its unmatched resolution, robustness, and scalability.[55,56] Continuous chromatography has been pursued by researchers due to the promise of higher productivity, higher purity, lesser operating cost, and lesser footprint.[5,57] Many different methods to achieve this have been proposed and are briefly described next.

Continuous Annular Chromatography

Continuous annular chromatography (CAC) contains an annular adsorbent bed packed between two concentric cylinders (Figure 4a). The feed is continuously introduced from a fixed point on the top of bed. Individual helical bands of each component are formed because of different elution rates of different components in the bed coupled with bed rotation. These helical bends extend to the bottom of the bed, and separated components are collected at different angular positions from the fed point. Hence CAC allows the operation of chromatographic purification in a truly continuous and steady-state manner. Also, CAC provides a major advantage

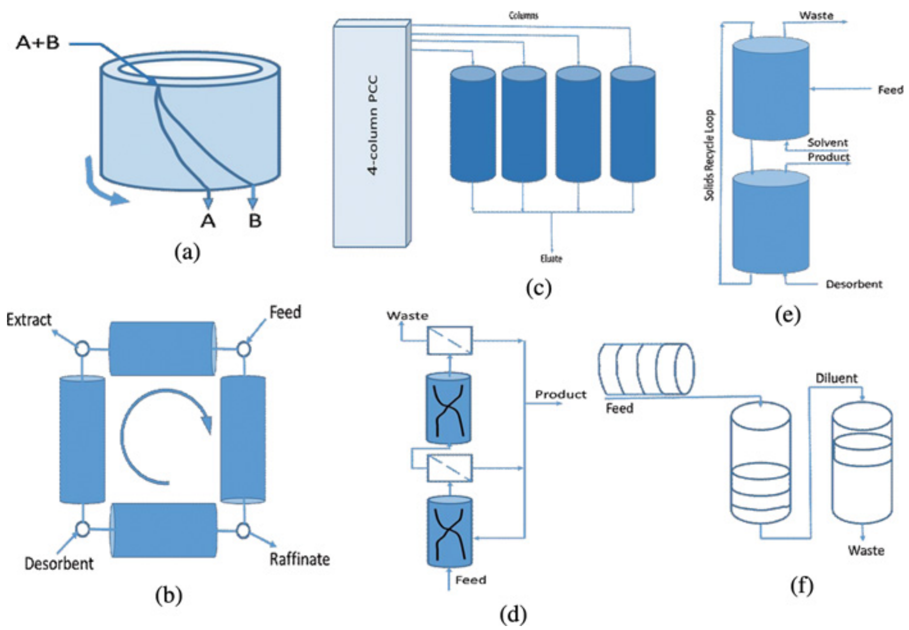

FIGURE 4 Continuous chromatography techniques: (a) continuous annular chromatography, (b) simulated moving bed chromatography, (c) periodic countercurrent chromatography, (d) expanded bed chromatography, (e) magnetically stabilized fluidized bed chromatography, and (f) multicolumn countercurrent solvent gradient purification chromatography.

in the form of straightforward transfer of step-elution protocol for batch chromatography to CAC column.[58] Some significant recent advances include implementation of stepwise elution (gradient elution when multiple concentrations are used) and displacement method.[59]

Simulated Moving Bed (SMB) Chromatography

SMB chromatography uses a series of chromatographic columns connected in series (Figure 4b). The positions of sample inlet and collection points are moved continuously to provide a simulation of true moving bed chromatography. SMB can therefore be used to provide high-resolution separation of compounds that otherwise have minor differences in their retention times. This is achieved via cumulative addition of these differences throughout the bed. Compared to theoretical plate-dependent separation, the costs associated with maintaining, operating, and obtaining larger single columns are reduced.[60] In this mode, roughly 50–70% of capacity of the affinity media is engaged in separation. SMB is presently used for high-resolution separation of binary mixtures such as chiral compounds.[61]

Countercurrent Chromatography (CCC)

In this format, multiple columns are used to run the different chromatographic steps (equilibration, regeneration, loading, and washing) discretely and continuously in a cyclic fashion

(Figure 4c). Since the columns are in series, the flow-through and wash from one column are captured by the second column. This unique feature allows for loading the resin close to its static binding capacity rather than dynamic binding capacity as in batch mode chromatography. An alternative method for performing CCC is magnetically stabilized fluidized bed chromatography.[2] It uses a fluidized bed that facilitates countercurrent solid/solvent contact (Figure 4d). The feed mixture is injected in the center of the bed and the desired product adheres to the affinity ligand while the nonadsorbed impurities travel countercurrently.[57] The solids are continuously removed from the bottom and transferred to the second MSFB, where a desorbent is used to remove the protein from the solids. The advantage that CCC has over SMB is that problems such as clogging due to debris and contamination can be minimized by treating the exiting solids.

Expanded Bed Chromatography (EBC)

Expanded bed chromatography (Figure 4e) permits expansion of the adsorbent bed due to upward movement of column fluid, thus allowing the passage of crude raw materials such as fermentation broth without clogging the column.[62] Affinity chromatography has been successfully used as expanded bed for isolation of proteins from cell broths. Expanded bed chromatography can be run in a continuous manner by using a series of contactors (countercurrent or cocurrent) in a loop for continuous loading, washing, elution, and reequilibration of the transported adsorbent. Continuous countercurrent expanded bed adsorption has been used successfully for direct purification of lysozyme.[63]

Multicolumn Countercurrent Solvent Gradient Purification Chromatography (MSCGP)

MSCGP is a continuous multicolumn chromatographic process that can perform high-resolution separation using linear, nonlinear, or segmented gradients (Figure 4f). The system is split into multiple columns, and each column performs the task analogous to batch chromatography. However, the countercurrent movement of the impure fractions through the column provides high yield and high purity simultaneously.[64]

Countercurrent Tangential (CCT) Chromatography

CCT combines the advantages of SMB and EBC formats. The resin is in the form of a slurry and flows through a series of static mixers and hollow fiber membrane modules.[65] During binding, the protein binds to the resin retained in the fiber membrane while impurities flow through the membrane and are removed as waste. Since the buffers used in the binding, washing, and elution steps flow countercurrently to the resin through multiple hollow fiber membranes, high-resolution separations can be achieved while increasing the product yield and reducing the amount of buffer needed for protein purification.[66]

Many other configurations of continuous chromatography have been reported in the literature. However, besides developing the different configurations, there is an increasing focus on novel ligands and matrix development so as to reduce the process time by overcoming mass transfer limitations due to diffusion. An example of the latter is the development of monolithic columns.[67,68]

Aqueous Two-Phase Extraction (ATPE)

Liquid–liquid extraction (LLE), such as aqueous two-phase extraction (ATPE), is a promising alternative to process chromatography and has been extensively examined in the last few decades.[69,70] In ATPE, the setup consists of three sections: a countercurrent extraction cascade, a back extraction stage, and a washing unit.[71] The system facilitates continuous separation and purification of the therapeutic protein. Also, ATPE makes it possible to integrate clarification and partial purification in a single step.

INTEGRATED BIOPROCESSING PLATFORM

The already-mentioned developments of unit operations conducive for achieving continuous bioprocessing have motivated the research community to focus on integration of individual innovations in order to create a continuous bioprocessing train.[72] With the large variety of strategies available in the literature for each unit operation, choosing the optimum strategy is critical. Effective decision making can yield significant benefits. For example, one can intelligently omit certain operations, thus improving the productivity and equipment requirement.

In one such study,[2] a continuous bioprocessing process train was demonstrated for production of biopharmaceuticals (Figure 5). This scheme utilized high-density perfusion culture through which continuously captured product was purified using a periodic countercurrent chromatography (PCC) system in an integrated manner. The scheme was implemented for production of monoclonal antibodies (mAbs) and Rh enzyme, and was tested for continuous operation for 30 days without any indication of performance deterioration. Equally importantly, the product quality was demonstrated to be comparable to the product manufactured by using a batch process. Similarly, a continuous process train was successfully developed recently for biosurfactant production.[73] The authors demonstrated that a significant part of biomass was immobilized onto the air/liquid membrane contactor. The process resulted in production of 11.5 g of surfactin after 3 days of culture. This translated to an average surfactin productivity of 54.7 mg L^{-1} hr^{-1} for the continuous culture and a surfactin productivity of 30 mg L^{-1} hr^{-1} at the steady state. Using the proposed integrated bioprocess, it was possible to continuously extract and purify several grams of biosurfactant with purity up to 95%.

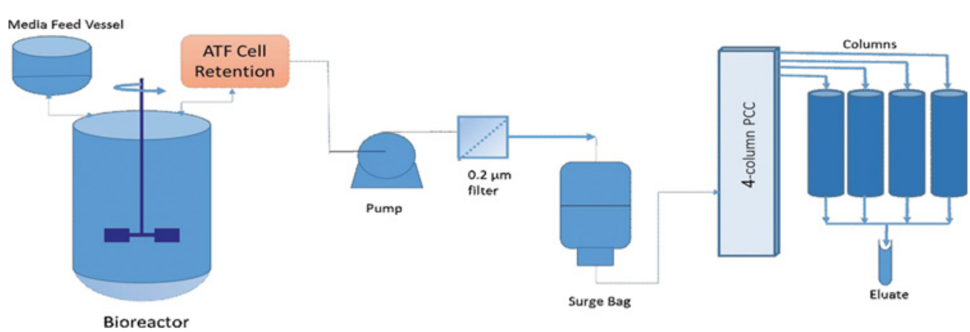

FIGURE 5 Continuous upstream processing bioreactor coupled with ATF provides continuous removal of product from bioreactor, combined with PCC to provide continuous purification.[2]

PATH FORWARD

Continuous bioprocessing has been hailed as the "facility of the future."[2] It has been a collective endeavor of the bioprocessing community to drive a paradigm shift toward continuous processing. The drivers include higher production, lesser capital investment, real-time process analysis, and improved quality control. Implementation of process analytical technology is likely to play a critical role in successful implementation of continuous processing. Also, real-time assessment of product quality attributes of continuously produced product ensures consistent quality. Since the entire process train runs in a closed and automated manner, there is higher robustness due to lesser chances of contamination and mixups and better compliance with current good manufacturing practices (cGMP).

Over the past decades, a number of investigations have contributed toward optimal and innovative designs of individual biotech unit operations for continuous operation. While improvisation at each step is very important, future research needs to focus on integration of the multiple unit operations into a single process[2] and optimization of the entire bioprocessing train as a whole.[40] Such studies would facilitate adoption of continuous bioprocessing in the bioprocessing industry.

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